

The status of the HOD conjecture

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Abstract

For the past 15 years, Woodin’s HOD conjecture has been a driving force in the study of large cardinal hypotheses. Although it is perfectly clear what it would mean to prove the conjecture, it is more subtle to say what it would take to refute it. That is not to say that the HOD conjecture is ill-defined, or that it fails to be a mathematical problem. This paper seeks to clarify the problem and its current status.

1 Introduction

Set theory currently faces an unprecedented development: the status of a major open problem hangs on a difference of opinion as to what constitutes evidence for the legitimacy of a large cardinal hypothesis. The question is whether Aguilera–Bagaria–Lücke’s recent work on exacting cardinals [3] constitutes a refutation of the HOD conjecture. Typically in mathematics, to know whether a conjecture has been refuted one simply checks whether the proof is correct; in this case, no one disputes that. The situation here is more complicated. Because the proof in question appeals to hypotheses that cannot be established from the traditional axioms of set theory, there remains the subtler issue of whether these hypotheses are justified. We will argue that they are not (at least not yet), and that the HOD conjecture remains an open question.

The HOD conjecture is a major open problem in the theory of large cardinals, and so it is to be expected that a solution to the HOD conjecture would make use of large cardinal axioms, set theoretic principles that posit the existence of very large sets (or equivalently very large cardinal numbers), sets so large that their existence cannot be proven from the usual axioms. The typical standpoint in the field is to adopt large cardinal axioms as true and try to determine their consequences.

Aguilera–Bagaria–Lücke [3] introduce the notion of an exacting cardinal, and argue that it is a natural large cardinal axiom. Then they show that the existence of exacting cardinals combined with accepted large cardinal axioms (for example, supercompact cardinals) refutes Woodin’s HOD conjecture.

If ZFC is consistent with an exacting cardinal above a supercompact cardinal, then Woodin’s HOD conjecture is false. This is an indisputable mathematical fact. So hasn’t the HOD conjecture been refuted, at least to the satisfaction of the large cardinal theorist?

We will argue that one aspect of the HOD conjecture has always been the contention that the sort of hypotheses underlying Aguilera–Bagaria–Lücke theorem (exacting cardinals and the like) are illegitimate. In fact, just a couple of pages after first formulating the HOD conjecture, Woodin explicitly conjectured that large cardinal hypotheses very similar to exacting cardinals are outright inconsistent. If he is correct, a refutation of the HOD conjecture from such hypotheses is vacuous. But regardless of whether these hypotheses are inconsistent,¹ to take Aguilera–Bagaria–Lücke’s work on its own as a refutation of the HOD conjecture amounts to begging the question.

What then would it take to refute the HOD conjecture? Is it even coherent for a set theorist to accept strong large cardinal hypotheses but conjecture the inconsistency of ultra-exacting cardinals? How can one decide which large cardinal hypotheses are legitimate? The bulk of the paper is devoted to these questions.

In §2 we motivate and state the HOD conjecture and discuss its implications for choiceless large cardinals. §3 summarizes the concepts and results from [3] which precipitated this article. This raises the question whether Woodin already refuted the HOD conjecture when he put it forth in [45], which is the topic of §4. We distinguish two kinds of potential refutations of the HOD conjecture in §5: definitive refutations and refutations from new axioms. We argue that a refutation from new axioms involves more than a theorem that a new axiom implies the HOD Hypothesis is false; it requires a mathematical case for the consistency of the new axiom. We argue in §6 that this is a pressing issue for (ultra)exacting cardinals. In §6.1 we show that the argument for the naturalness and consistency of the new axiom in the Aguilera–Bagaria–Lücke theorem is invalid in general. We define *γ -stationarily exacting cardinals* in terms of structural reflection and prove their existence is consistent relative to axiom I_2 . We then show that a γ -stationarily exacting cardinal

¹In fact, we do not believe this.

above an extendible cardinal is inconsistent. In §7 we describe ways that the consistency of choiceless cardinals may eventually be certified. If this were to happen, we argue that it would not spell the end of inner model theory, contrary to [6]. It would be one of inner model theory’s greatest successes.

2 The HOD conjecture

The HOD conjecture, in one formulation, asserts that it is provable from large cardinal axioms that the class $\text{Ord}^{\mathbb{N}}$ of infinite sequences of ordinals has a well-order that is definable from parameters over the structure $(V, \in)$. To make the conjecture precise, one can pick a specific large cardinal axiom (the reason being that “large cardinal axiom” is not a precisely defined notion). The official hypothesis is a supercompact cardinal, but a proof of the HOD conjecture from any of the standard large cardinal hypotheses, no matter how strong, would constitute a major advance in set theory.

The conjecture was originally formulated (in a different but equivalent form) by Hugh Woodin, who himself acknowledged its counterintuitive nature.² The axiom of choice implies that any set can be well-ordered, but one typically does not expect these well-orderings to be definable. For example, it is consistent that there is no well-ordering of the Baire space $\mathbb{N}^{\mathbb{N}}$ that is definable without parameters in the structure $(V, \in)$.³ Of course, there is a well-ordering of this space that is definable from parameters, since one can use the well-ordering itself as a parameter. The required parameter is a subset of $\mathbb{N}^{\mathbb{N}}$. Letting $\theta = (2^{\aleph_0})^+$, it is consistent that there is no well-ordering of $\theta^{\mathbb{N}}$ that is definable from a subset of $\mathbb{N}^{\mathbb{N}}$. In general, one would expect that as κ increases, definably well-ordering $\kappa^{\mathbb{N}}$ requires larger and larger parameters. According to the HOD conjecture, large cardinal axioms imply that this is not the case.

In light of the HOD conjecture’s counterintuitive nature, one might ask why Woodin conjectured it at all. The motivation for the HOD conjecture comes from a series of technical results in large cardinal theory and inner model theory.

²Early on, Woodin referred to it as “the silly conjecture” [6, p. 305].

³In other words, it is consistent that $\mathbb{R} \notin \text{HOD}$.

2.1 Why make the HOD conjecture?

A set is *ordinal* definable if it is the unique solution to a first order formula allowed finitely many ordinal parameters. Gödel introduced the transitive universe of hereditarily ordinal definable sets (HOD). It consists of those ordinal definable sets whose transitive closure is ordinal definable. It is an inner model of any universe of set theory, and it is a natural model of the axiom of choice.

As Kunen puts it, “Most interesting questions about HOD have answers which are independent of ZFC” [28, p. 162]. This includes the question of whether every set belongs to HOD, or in other words, whether the universe of sets V is equal to HOD. None of the standard large cardinal axioms implies that $V \neq \text{HOD}$, but it is consistent relative to large cardinals that HOD be much smaller than V . To take an extreme example, it is consistent for there to be a proper class of supercompact cardinals in V but none in HOD [10]. It is consistent relative to large cardinals that HOD not compute successor cardinals correctly [11]. Such consistency results comprise another reason why the HOD conjecture seems, to the set theorist, unlikely.

Solovay’s Splitting Theorem says that any stationary subset of a regular cardinal κ can be partitioned into κ -many stationary sets. A regular uncountable cardinal κ is *ω -strongly measurable in HOD* if there is $\gamma < \kappa$ such that

1. $(2^\gamma)^{\text{HOD}} < \kappa$, and
2. the set of points below κ of countable cofinality cannot be definably partitioned into γ -many sets.

In other words, κ is *ω -strongly measurable in HOD* if there is no witness in HOD to Solovay’s Splitting Theorem for the points in κ of countable cofinality.

Large cardinals are most often used to show relative consistency results like those about HOD just described. The next theorem is surprising because it says that large cardinals *imply* the universe of sets is either “almost HOD” or not at all approximated by HOD.

Theorem 2.1 (HOD Dichotomy Theorem; Woodin). *Suppose κ is an extendible cardinal. Then exactly one of the following holds.*

1. *No regular cardinal above κ is ω -strongly measurable in HOD.*
2. *All regular cardinals above κ are ω -strongly measurable in HOD.*

The *HOD Hypothesis* states that case (1) holds: the set of points of countable cofinality below any sufficiently large regular cardinal can be definably partitioned into stationary sets.

Definition 2.1 (HOD Hypothesis). There are arbitrarily large regular cardinals which are not ω -strongly measurable in HOD.

The HOD conjecture is that the HOD Hypothesis is provable from large cardinals which suffice to prove the HOD Dichotomy.⁴

Conjecture 2.2 (The HOD conjecture). There is a proof of the HOD Hypothesis in the theory $\text{ZFC} + \text{“there is a supercompact cardinal.”}$

The main motivations for the HOD conjecture require some background to explain. As a first pass, large cardinal hypotheses are strong versions of the axiom of infinity which imply the consistency of ZFC and much stronger theories. By Gödel’s second incompleteness theorem, ZFC cannot prove large cardinal hypotheses. If ZFC embodies the accepted first principles of mathematics, then what reason is there for believing large cardinal hypotheses are true? Or even consistent?

Inner model theory is the best possible answer to the consistency question. Given a large cardinal hypotheses, inner model theorists aim to solve its *inner model problem*: the problem of constructing a fine structural inner model in which that large cardinal hypothesis is satisfied. *Fine structural* means there is a definability analysis of the model which can in principle be leveraged to answer effectively any question which can be asked about it. (Because they are definable, fine structural inner models are contained in HOD.) If a large cardinal is inconsistent, the fine structure theory of a supposed inner model for it will easily yield a contradiction.

Once a large cardinal hypothesis is certified by an inner model, the consistency of any theory of the large cardinal’s consistency strength is likewise certified; one can force over the inner model to produce a model of the theory. Thus inner models for large cardinals certify entire consistency strengths.

Just as important as the “pure” part of inner model theory—the *Inner Model Program* to build fine structural models for every large cardinal hypothesis—is its “applied” part. This is *core model theory*, the measurement of consistency strength lower bounds of strong theories using fine structural inner models.

⁴The second author proved the HOD Dichotomy Theorem from a strongly compact cardinal [21].

It is a reasonable position to trust large cardinal hypotheses whose inner model problems have been solved and to be more reserved about those which inner model theory has not yet reached. How far into the large cardinal hierarchy should such a person's trust extend? Inner model theory proceeds incrementally, from weaker to stronger large cardinal hypotheses. Inner models are targeted precisely to their large cardinal hypotheses: the inner model for one measurable cardinal does not contain two measurable cardinals, or a strong cardinal. Inner model theory presently reaches large cardinal hypotheses up to the hypothesis “there is a Woodin cardinal that is a limit of Woodin cardinals” by a construction due to Neeman [33]. Getting beyond this hypothesis is an open problem.

The difficulty of inner model theory is contrary to early optimism. It was expected in the 1970s that an inner model for a supercompact cardinal was forthcoming. Solving the inner model problem for a supercompact is now one of the foremost open problems in set theory, and the gap in the large cardinal hierarchy between where inner model theory reaches and a supercompact is substantial. In order to understand this difficulty, in the mid-2000s, Woodin began investigating what inner model theory for a supercompact would have to look like. This led him to isolate the notion of a *weak extender model*, which is an inner model that inherits a supercompact cardinal from the universe of sets in a natural way.

The possibility that the solution to the inner model problem for a supercompact is a weak extender model is particularly appealing because of Woodin's Universality Theorem. It says that if there is a weak extender model, then that model inherits all extenders witnessing large cardinal properties in V . So the inner model for a supercompact would also be an inner models for essentially all stronger large cardinal axioms (provided those axioms hold in V). Inner model theory would no longer be a Sysiphean task; it would in a sense be completed.

What does this have to do with the HOD conjecture?

Theorem 2.2 (Woodin [46]). *If there is an extendible cardinal δ , then the following are equivalent.*

1. *HOD is a weak extender model for the supercompactness of δ .*
2. *Some regular cardinal greater than δ is not ω -strongly measurable in HOD.*

The HOD Dichotomy may then be recast in terms of weak extender models: If there is an extendible cardinal δ , then either HOD is a weak extender model in

which δ is supercompact, or else not only is HOD not a weak extender model for the supercompactness of any $\kappa \geq \delta$, but HOD contains no weak extender models witnessing supercompactness either [46].

Woodin cites this formulation and the abstract possibility of inner model theory for supercompact cardinals as motivation for making the HOD conjecture [46, p. 27]. The conjecture that this inner model theory is realizable in the form of a weak extender model satisfying $V = \text{HOD}$ is the Ultimate L conjecture.

Conjecture 2.3 (Ultimate L conjecture). If there is an extendible cardinal δ , then there is a weak extender model for the supercompactness of δ which satisfies the axiom $V = \text{Ultimate } L$ and is contained in HOD.

Clearly the Ultimate L conjecture implies the HOD Hypothesis.

2.2 The HOD conjecture & choiceless cardinals

The HOD conjecture has striking implications for *choiceless cardinals*, large cardinals so large they contradict the axiom of choice. A cardinal κ is a *Reinhardt cardinal* if κ is the critical point of a nontrivial elementary embedding $j : V \rightarrow V$.

Theorem 2.3 (Kunen Inconsistency Theorem). *Assume ZFC. There are no Reinhardt cardinals. In fact, there is no nontrivial elementary embedding $j : V_{\lambda+2} \rightarrow V_{\lambda+2}$.*

The Kunen Inconsistency was surprising to the community. Jensen writes

At first, Reinhardt cardinals were greeted by many set theorists, including the author of this article, as the embodiment of an intuitively especially satisfactory existence principle. But after a few years Kunen proved that these cardinals do not exist. This set an upper limit for the hierarchy. [25, p. 399]

Kunen’s argument uses the axiom of choice. The question whether Reinhardt cardinals are refutable in ZF was immediately raised, the authors of [36, p. 83] writing that “Pending an answer to [this question], one can perhaps best view [the Kunen Inconsistency] as a structural limitation imposed on V by the Axiom of Choice.” Attempts at refuting Reinhardt cardinals in ZF have resulted in multiple new proofs of Kunen’s theorem, none of which avoid choice.

Perhaps the reason for this is that ZF is not powerful enough to prove *any* nontrivial large cardinal-like hypothesis is inconsistent. In the 1990s, Woodin began formulating hierarchies of stronger and stronger choiceless cardinals—large cardinal-like principles which are incompatible with the axiom of choice—to test this. One hierarchy is the *Reinhardt hierarchy*.

Definition 2.4 (Reinhardt hierarchy). A cardinal κ is

1. *Reinhardt* if κ is the critical point of an elementary embedding from V to itself.
2. *super Reinhardt* if for all α there is an elementary embedding from V to itself with critical point κ and such that $j(\kappa) > \alpha$.
3. *totally Reinhardt* if for each $A \subseteq V_\kappa$,

$$\langle V_\kappa, \wp(V_\kappa) \rangle \models \text{ZF}_2 + \text{“there is a cardinal which is super Reinhardt for } A\text{.”}^5$$

The other hierarchy is the *Berkeley hierarchy*.

Definition 2.5 (Berkeley hierarchy). A cardinal κ is

1. *Berkeley* if for all transitive sets M with $\kappa \in M$ and every ordinal $\eta < \kappa$, there is an elementary embedding $j : M \rightarrow M$ whose critical point is between η and κ .
2. *club Berkeley* if κ is regular and for any club $C \subseteq \kappa$ and any transitive set M with $\kappa \in M$, there is an elementary embedding from M to itself whose critical point is in C .
3. *limit club Berkeley* if κ is a club Berkeley cardinal and a limit of Berkeley cardinals.

Berkeley cardinals are refutable in ZFC with a simpler proof than Kunen’s refutation of Reinhardt cardinals. This may be seen as evidence for their being inconsistent in ZF. On the other hand, the theory of Berkeley cardinals is coherent modulo independence. For instance, if there is a Berkeley cardinal, the amount of choice that holds, as measured by generalizations of Dependent Choice, is determined by

⁵If A is a proper class, then a cardinal κ is *super Reinhardt for A* if κ is super Reinhardt as witnessed by embeddings which fix A .

the cofinality of the least Berkeley cardinal. (It is open whether the least Berkeley cardinal can be regular.)

The Reinhardt hierarchy is proper. This is proved using reflection arguments [6]. Within the Berkeley hierarchy, it is not known exactly how club Berkeley and Berkeley cardinals relate; in particular, it is open whether club Berkeley cardinals *rank-reflect* Berkeley cardinals [6, Question 4]. But a limit club Berkeley cardinal is a club Berkeley cardinal, and any club Berkeley cardinal is a totally Reinhardt cardinal. So the Berkeley hierarchy eventually extends the Reinhardt hierarchy.

None of these choiceless cardinals has been refuted in ZF. On the standard picture that choiceless cardinals continue the large cardinal hierarchy past the large cardinals compatible with choice, a large cardinal set theorist committed to studying consistency strengths may have only one way up the consistency strength hierarchy, and it may require giving up the axiom of choice. For this reason, combined with the expectation that an inconsistency proof was just around the corner, choiceless cardinals have been viewed as deviant notions.

The widespread acceptance of choice among set theorists rests on pragmatic grounds. There remains (for some of the more philosophically inclined) the question *why* it is true. The HOD conjecture arguably provides that justification, since it says that assuming large cardinal axioms, but not assuming the axiom of choice, V is provably “close” to a natural model of choice.

If true, the HOD conjecture would do much more to bolster choice. It would provide the ZF refutation of much of the choiceless cardinal hierarchy. It would say there is no consistency strength inaccessible to choice.

Theorem 2.4 (Woodin). *Suppose that ZF is consistent with the existence of a Reinhardt cardinal and arbitrarily large supercompact cardinals. Then the HOD conjecture is false.*

If the HOD conjecture is true, there cannot be a Reinhardt cardinal with an ω -huge cardinal greater than $\kappa_\omega(j)$, and Berkeley cardinals cannot exist [6]. In a sense, Kunen’s Inconsistency would be optimal:

Theorem 2.5 (Woodin [46]). *If the HOD conjecture is true, then it is provable in ZF that if δ is an extendible cardinal, then for all $\lambda \geq \delta$, there is no nontrivial elementary embedding $j : V_{\lambda+2} \rightarrow V_{\lambda+2}$.*

So assuming the HOD conjecture, certain configurations of choiceless and accepted large cardinal hypotheses are ruled out. Theorems 2.4 and 2.5 leave open

that the existence of Reinhardt cardinal may be a consistent hypothesis, assuming the HOD conjecture. If it is, and the HOD conjecture holds, then certain conjunctions of individually consistent large cardinal hypotheses are not jointly consistent. If there is a Reinhardt cardinal, there can be no extendible cardinals below it.

Assuming certain choiceless hypotheses—like Berkeley cardinals, or a Reinhardt with a proper class of supercompact cardinals—are consistent, then they stand in relation to HOD as measurable cardinals stand in relation to the Constructible Universe. They imply that the universe is not HOD. They serve as “HOD[#]”.

In this connection, [6] describes “two futures” according to whether the HOD conjecture is true or not. The future in which it is true is described as one in which inner model theory “succeeds.” The future in which it is false and choiceless cardinals are consistent is a future in which inner model theory “fails.”

If large cardinals beyond choice are *consistent*, then, as we have seen, the Weak HOD Conjecture fails, and hence the Ultimate- L Conjecture fails, and this constitutes an anti-inner model theorem. We should stress that the result isn’t about just a single inner model—Ultimate- L . It is much more general. Inner model theory proceeds in a general setting, by assuming a large cardinal hypothesis and then showing that one can build a canonical inner model for that large cardinal. The failure of the Weak HOD Conjecture would show that even if one assumes that there is an extendible cardinal κ with a huge cardinal above it (or even much stronger large cardinal hypotheses) then one cannot show that there is a weak extender model N of the supercompactness of κ such that $N \models “V = \text{HOD}”$. It is hard to see what meaning there could be to inner model theory if it cannot be executed even at this level of generality. So, in this scenario, not only would the Weak Ultimate- L Conjecture be false, but we would also very likely be faced [with] the failure of inner model theory and the failure of the prospect of ever having detailed, fine-structural insight into V . This is the future in which *chaos* prevails. [6, p. 315]

We will return to this below.

2.2.1 ZFC versions of choiceless cardinals

Versions of choiceless cardinals compatible with choice can be obtained by relativizing them to pertain to inner models of ZFC rather than V . Let N be an inner model of ZFC. Then, for example, a cardinal κ is *N-Reinhardt* if it is the critical point of an elementary embedding from N to N , and κ is *N-Berkeley* if for all transitive $M \in N$ with $\kappa \in M$, and for all $\eta < \kappa$, there is a non-trivial elementary embedding $j : M \rightarrow M$ with critical point between η and κ . In this language, $0^\#$ exists just in case there is an L -Reinhardt cardinal.

The HOD-analogues of choiceless cardinals (e.g., HOD-Reinhardt and HOD-Berkeley cardinals) can be viewed as analogs of $0^\#$ for HOD and provide another avenue for refuting the HOD conjecture. If there is a HOD-Berkeley cardinal, the HOD Conjecture is false [6, Theorem 8.7].

3 Ultra-exacting cardinals

The first proof of Projective Determinacy was Woodin’s proof from infinitely many Reinhardt cardinals. Bearing the Kunen Inconsistency in mind, the significance of the theorem was unclear. To obtain a meaningful theorem, Woodin weakened the hypothesis to a large cardinal hypothesis consistent (as far as we know) with the axiom of choice, thereby isolating the axiom I_0 .⁶

The consistency of a Berkeley cardinal implies the HOD conjecture is false. In the presence of a proper class of supercompact cardinals, so does the consistency of a *rank-Berkeley* cardinal, a choiceless weakening of Berkeley cardinals introduced in [23]. Aguilera-Bagaria-Lücke weaken the definition of rank-Berkeley cardinal to arrive at hypotheses consistent with choice. The resulting hypotheses are *exacting* and *ultraexacting* cardinals.

Theorem 3.1 (Aguilera-Bagaria-Lücke [3]). *Suppose the theory $\text{ZFC} + \text{“there is an exacting cardinal above an extendible cardinal”}$ is consistent. Then the HOD Hypothesis is false.*

The question of whether one should believe the consistency of the hypothesis of Theorem 3.1 is subtle. If $\text{ZFC} + I_0$ is consistent, then so is $\text{ZFC} + \text{“there is an$

⁶ $I_0(\lambda)$ is the assertion that there is an elementary embedding $j : L(V_{\lambda+1}) \rightarrow L(V_{\lambda+1})$ with critical point below λ , and I_0 asserts that for some λ , $I_0(\lambda)$ holds. This axiom is stronger than earlier discovered rank-to-rank embedding axioms I_3 , I_2 and I_1 . The “I” in these names stands for “(probably) inconsistent,” as the axioms were suspected of being too much like Reinhardt’s axiom.

ultraexacting cardinal” [5, Theorem C]. Also $\text{ZFC} + I_0$ implies the consistency of extendible cardinals. Using these two facts, Aguilera–Bagaria–Lucke have argued that it is not such a leap to think that an extendible below an exacting cardinal is consistent. On the other hand, the consistency strength of an ultraexacting above an extendible cardinal is beyond I_0 , and the consistency of an exacting cardinal above an extendible cardinal has only been shown consistent relative to a choiceless hypothesis.

Theorem 3.2 (Aguilera–Bagaria–Lücke [3]). *The theory $\text{ZF} +$ “there is a $C^{(3)}$ -Reinhardt cardinal with a supercompact cardinal above the supremum of its critical sequence” proves the consistency of the theory $\text{ZFC} +$ “there is an ultraexact cardinal which is a limit of extendible cardinals.”*

4 Did Woodin refute the HOD conjecture at the same time he made it?

The first published statement of the HOD conjecture appears in Woodin’s “Suitable Extender Models I” [45, Definition 191]. Shortly after stating the conjecture, Woodin proves the following theorem:

Theorem 4.1 (Woodin). *Assume the HOD conjecture. Suppose there is an HOD-supercompact cardinal below λ . Then there is no non-trivial elementary embedding $j : V_{\lambda+1} \rightarrow V_{\lambda+1}$ such that for all Σ_2 -formulas $\phi(x)$, for all $a \in V_{\lambda+1}$, $V \models \phi[a]$ if and only if $V \models \phi[j(a)]$.*

Let us denote by $W(\lambda)$ the hypothesis that there is a non-trivial elementary embedding $j : V_{\lambda+1} \rightarrow V_{\lambda+1}$ such that for all Σ_2 -formulas $\phi(x)$, for all $a \in V_{\lambda+1}$, $V \models \phi[a]$ if and only if $V \models \phi[j(a)]$. The contrapositive to Woodin’s theorem above then reads: if there is an HOD-supercompact cardinal below λ and $W(\lambda)$ holds, then the HOD hypothesis fails. Woodin noted that this shows that the HOD hypothesis can be refuted from something that looks superficially like a large cardinal axiom:

Should the HOD conjecture be consistently false, an obvious question is what large cardinal hypothesis (consistent with the Axiom of Choice) might suffice to establish this. The following theorem⁷ . . . does indicate a

⁷Woodin refers here essentially to the contrapositive of Theorem 4.1

sufficient hypothesis but at present there is no basis from which one can argue that this is a large cardinal hypothesis. Whether this approach can lead to a genuine large cardinal hypothesis which is also sufficient is far from clear. [45, p. 313]

The notion of a “genuine large cardinal hypothesis” at play in this cryptic comment is far from clear, and we will attempt to make sense of it below.

One might think that Aguilera-Bagaria-Lücke’s contribution is to supply a genuine large cardinal hypothesis that refutes the HOD conjecture; namely, an (ultra)exacting cardinal. However, the following result seems to imply that ultraexacting cardinals are no more genuine than Woodin’s original hypothesis.

Proposition 4.2 (Aguilera–Bagaria–Lücke–Goldberg). *The following are equivalent:*

- (1) λ is ultra-exacting.
- (2) $W(\lambda)$.

Proposition 4.2 is the basis for our contention that the conjecture that the hypotheses of Theorem 3.1 are inconsistent has been an implicit underlying assumption of the HOD conjecture from its inception. In fact, Woodin explicitly makes an even stronger inconsistency conjecture:

A natural conjecture is that for all λ , if there is a supercompact cardinal below λ then there is no non-trivial elementary embedding

$$j : V_{\lambda+1} \rightarrow V_{\lambda+1}$$

such that for all formulas $\varphi(x)$, for all $a \in V_{\lambda+1}$,

$$V_{\lambda+2} \models \varphi[a]$$

if and only if $V_{\lambda+2} \models \varphi[j(a)]$. [45, p. 304]

The hypothesis of Woodin’s first proof of PD may turn out to be vacuous, so he did not take it to be the definitive answer to the question whether large cardinals imply PD. Similarly, he believed the hypothesis of Theorem 4.1 is vacuous. Solving the HOD conjecture via this route requires validating the consistency of an

ultraexacting cardinal (or equivalently $W(\lambda)$) with a HOD-supercompact below it or reducing to a safe hypothesis as he did in the case of PD. So it was rational for Woodin not to believe he had refuted the conjecture immediately after making it.

5 What does it take to solve the HOD conjecture?

In this section, we describe several forms that a solution to the HOD conjecture might take.

Obviously, if the HOD conjecture is true, its solution consists of a proof of the HOD hypothesis in the theory $\text{ZFC} + \text{“there is a supercompact cardinal.”}$ There is nothing mysterious about what it would take to prove the HOD conjecture.

On the other hand, the negation of the HOD conjecture states that the theory $\text{ZFC} + \text{“there is a supercompact cardinal”} + \text{the negation of the HOD hypothesis}$ is consistent. What could a refutation of the HOD conjecture look like? It has to be a proof, and clearly the theory used in the proof has to be at least as strong in consistency strength as $\text{ZFC} + \text{“there is a supercompact cardinal.”}$ Perhaps the negation of the HOD hypothesis has very high consistency strength. In that case a proof of the negation of the HOD conjecture would have to use a very strong theory. In light of the “implicit conjecture” articulated above, this is the point at which what counts as a refutation becomes a subtle issue.

Suppose for illustration that the PA *conjecture* is the assertion that PA is inconsistent. Is a proof of the consistency of PA in second order arithmetic a refutation of the PA conjecture? If one believed PA is inconsistent, one is unlikely to be convinced otherwise by a proof in a theory of greater consistency strength.

When it comes to the HOD conjecture, there are ample resources stronger than $\text{ZFC} + \text{“there is a supercompact cardinal”}$ available for convincingly refuting it. What complicates the picture is that there are also resources that imply the HOD Hypothesis fails but which are, at present, akin to using second order arithmetic in a refutation of the PA conjecture. These resources are those “implicitly conjectured” to be inconsistent. We discuss refutations using each type of resource below.

5.1 Definitive refutations

Let us say that a *definitive refutation* of the HOD conjecture is a proof from standard large cardinal hypotheses that the HOD Hypothesis is false. A “standard” large cardinal hypothesis, for present purposes, is a large cardinal hypothesis about which

there is agreement that it is a large cardinal notion. A marker of this may be that the hypothesis appears in Kanamori's book [26]. The large cardinal used in a definitive refutation has to be at least as strong as a supercompact cardinal, consistency-wise. A very strong definitive refutation of the HOD conjecture would be a proof that if ZFC is consistent with a supercompact cardinal, then so is ZFC plus a supercompact cardinal plus arbitrarily large regular cardinals that are ω -strongly measurable in HOD.⁸

Perhaps the “most definitive” conceivable refutation would be to show that the HOD Hypothesis can be forced to fail preserving all large cardinals, a sort of Lévy-Solovay theorem for the HOD Hypothesis.

5.1.1 The Strong HOD conjecture

When Woodin first formulated the HOD conjecture [45], there was no known model of ZFC in which more than three regular cardinals were ω -strongly measurable in HOD. This fact was another reason Woodin made the HOD conjecture, and it supports an even stronger conjecture.

Conjecture 5.1 (Strong HOD conjecture). *ZFC proves the HOD Hypothesis.*

Ben-Neria and Hayut [7] showed that the existence of a proper class of ω -strongly measurables in HOD is not actually such a strong hypothesis:

Theorem 5.1 (Ben-Neria–Hayut). *If there is an inaccessible cardinal κ such that $\kappa = \sup_{\alpha < \kappa} o(\alpha)$, then it is consistent with ZFC that every successor of a regular cardinal is ω -strongly measurable in HOD.*

The Ben-Neria–Hayut theorem suggests that the only route to proving the HOD hypothesis in ZFC would go through the combinatorics of successors of singular cardinals. It remains open, however, whether there can be more than one successor of a singular cardinal that is ω -strongly measurable in HOD. It is also open whether the successor of a singular cardinal of uncountable cofinality can ever be ω -strongly measurable in HOD. So there are several remaining avenues by which one could conceivably establish the HOD Hypothesis in ZFC.

Ben-Neria and Hayut's theorem answered the question whether there can exist more than three regular cardinals which are ω -strongly measurable in HOD, but it

⁸It seems likely that this is impossible, and instead the latter theory proves the consistency of the former, but this is open.

does not shed light on the HOD conjecture, since their technique can never produce a model of ZFC containing a supercompact cardinal. The existence of ω -strongly measurable cardinals is most interesting—and bears directly on the HOD conjecture—in the presence of the large cardinals at which the HOD Dichotomy kicks in.

Answering a question of Ben-Neria and Hayut, Aksornthong, Gappo, Holland, and Sargsyan obtained a property stronger than the ω -strong measurability of ω_1, ω_2 , and ω_3 by forcing over a determinacy model [4]. Whether this line of research can eventuate in a refutation of the HOD conjecture depends on whether there can exist a supercompact cardinal in a ZFC forcing extension of a model of determinacy.

5.2 Refutations from new axioms

An *refutation from new axioms (RNA)* of the HOD conjecture consists of

1. a proposal of a new purported large cardinal notion,
2. a proof that the proposed principle implies the HOD Hypothesis is false, and
3. a convincing case that the principle is consistent.

An RNA may seem like a departure from the normal standards of proof and refutation in mathematics. Usually a solution to a mathematical problem does not require extending the very axiomatic framework of mathematics further up the consistency strength hierarchy in a rigorously justifiable way. But on one construal, set theorists are already familiar with RNAs of other (tacit) conjectures. Let us explain.

5.2.1 The L Dichotomy and $0^\#$

The HOD Dichotomy Theorem is essentially a *covering* theorem for HOD analogous to Jensen’s Covering Theorem for the Constructible Universe L . Woodin often recasts Jensen’s theorem as the *L Dichotomy Theorem* to stress this similarity [46]. Each theorem says that either V is closely approximated by the respective inner model or else the inner model is radically wrong about cardinal structure.

Theorem 5.2 (*L Dichotomy Theorem; Jensen*). *Exactly one of the following holds.*

1. (*L hypothesis*) *Every uncountable set of ordinals is covered by a constructible set of the same cardinality.*

2. ($0^\#$ exists) *Every uncountable cardinal is an inaccessible cardinal in L .*

This should be contrasted with the following formulation of the HOD dichotomy theorem:

Theorem 5.3 (HOD Dichotomy Theorem; Woodin). *If κ is extendible, exactly one of the following holds.*

1. (HOD hypothesis) *Every set of ordinals of size at least κ is covered by an ordinal definable set of the same cardinality.*
2. *Every regular cardinal greater than or equal to κ is measurable in HOD.*

The L conjecture states that ZFC proves the L hypothesis. There is a general consensus in set theory that the L conjecture has been refuted by new axioms. The relevant large cardinal notion is the existence of a measurable cardinal; the proof that the L hypothesis is false from a measurable cardinal is essentially due to Silver [35]; and the case for the consistency of measurable cardinals draws on the work of many researchers including Harrington, Jensen, Kunen, Martin, Scott, Silver,⁹ and Solovay.

Notably, Silver’s theorem that measurable cardinals refute the L conjecture does not stand alone as an RNA of the L conjecture. On the state of knowledge at the time of his proof, one might instead have interpreted Silver’s theorem as evidence for the inconsistency of measurable cardinals. In fact, the philosophical case for the consistency of measurable cardinals is much more mathematically involved than Silver’s proof and rests on subsequent developments in the theory of $0^\#$ and the inner model theory of measurable cardinals.

Silver isolated $0^\#$ as the theory of a club class of indiscernibles for L , and proved remarkable global theorems for the structure of L under the hypothesis that $0^\#$ exists. As Maddy writes, “The point is that $0^\#$ yields a rich explanatory theory of exactly where and why L goes wrong. Before Silver, many mathematicians believed that $V \neq L$, but after Silver they knew why” [30, p. 506]. Solovay studied the descriptive set theoretic properties of $0^\#$, proving that in a sense it is the simplest possible set of numbers that is not constructible. Kunen showed that the existence of $0^\#$ is equivalent to there being a non-trivial elementary embedding from L to L

⁹Silver is a notable example of a set theorist who did not buy the case for the consistency of measurable cardinals [31].

(i.e. to there being an L -Reinhardt cardinal). Jensen’s Covering Theorem pinpointed the existence of $0^\#$ as a necessary and sufficient condition for “transcending” L .

Why believe there is a non-trivial elementary embedding from L to itself? The existence of $0^\#$ and measurability have descriptive set theoretic consequences. Solovay showed that measurability implies the regularity properties for Σ_2^1 sets. Combined work of Martin and Harrington shows that $0^\#$ exists if and only if lightface analytic games are determined, and Martin showed that if there is a measurable cardinal, then all analytic games are determined. These “low level” consequences of $0^\#$ and measurability gave set theorists a grasp of what set theory beyond L is like and started the development of descriptive set theory beyond ZFC.

The “high level” consequences of these hypotheses were then studied in $L[U]$, the inner model for a single measurable cardinal. Solovay defined $L[U]$ and outlined its fine structure. Silver showed that it satisfies the Generalized Continuum Hypothesis and has a Δ_3^1 -wellordering of its reals. Kunen showed that $L[U]$ does not depend on which κ -complete ultrafilter on κ the model is constructed from and that if $L[U]$ and $L[W]$ are inner models witnessing the measurability of κ_1 and κ_2 , then one of $L[U]$ and $L[W]$ is an iterated ultrapower of the other. These facts imply that the theory of $L[U]$ is invariant or *canonical*. It was immediately seen how to generalize the $L[U]$ construction to obtain models with many measurable cardinals and how to generalize $0^\#$ to $0^\dagger$, the sharp for $L[U]$.

Measurable cardinals became entrenched in set theoretic research. The first connection between determinacy and large cardinals was Solovay’s theorem that AD implies ω_1^V is measurable in HOD. Using core model theory, it is easy to see that most theories extending ZFC and studied intensively by set theorists imply the consistency of measurables. Understanding those theories requires the consistency of measurable cardinals. For example, using Mitchell’s generalization of the Covering Lemma, Gitik showed that the negation of the Singular Cardinals Hypothesis is equiconsistent with a measurable cardinal κ with $o(\kappa) = \kappa^{++}$.

We also have detailed understanding of what the universe of sets would look like if $L[U]$ did not exist via the Dodd-Jensen core model K^{DJ} , the largest fine structural inner model assuming the *anti-large cardinal hypothesis* that there is no inner model with a measurable cardinal.

These results—especially the definition and theory of $L[U]$ —collectively justify the consistency of the theory ZFC + “there is a measurable cardinal.” (It is telling that measurable cardinals were first investigated by Ulam in the 1930s but they only

acquired the status of a routine working hypothesis after $L[U]$ was defined.) Having them in hand, Silver’s theorem is generally accepted as a definitive RNA of the L conjecture.

It must be emphasized how critical $L[U]$ is to the case. A few years before Silver’s results, Scott proved that if there is a measurable cardinal, then the Axiom of Constructibility is false. If the L conjecture had been made, what would have been made of Silver’s theorem after Scott’s? It uses a resource which the L conjecture implies, via Scott’s theorem, is illegitimate. The strongest evidence that $V = L$ is false is the theory of $L[U]$ and the theory of $0^\#$. We understand each theory as clearly as we understand that of L .

5.2.2 Potential RNAs of the HOD conjecture

Returning to HOD, there are now two potential RNAs of the HOD conjecture.

Large cardinals beyond choice. One path towards an RNA of the HOD conjecture, suggested by Bagaria–Koellner–Woodin [6], is to argue for the consistency of large cardinals beyond choice. There is a growing body of evidence of the consistency of these cardinals, mostly in the form of structure theory [12], [23], [20], [18].

We emphasize that the question of whether the Axiom of Choice is *true* is at least superficially irrelevant to the question of mounting an RNA of the HOD conjecture via choiceless cardinals. This is because one only has to argue for their consistency. However, in many ways the theory of choiceless cardinals remains mysterious precisely because we must give up choice: though the large cardinals themselves lead to a rich structure, ZF is a very weak background theory and it is unclear how to strengthen it to clarify the picture. (Due to the independence phenomenon, ZFC arguably already suffers from such a lack of clarity, but at least in that case, thanks to the massive development of forcing, we know for the most part what we cannot know.)

Section 7.1 suggests another route to securing the consistency of choiceless cardinals that seems to go beyond mere structure theory: generalizing Schlutzenberg’s proof of the consistency of elementary embeddings from $V_{\lambda+2}$ to itself.

The case of ultraexacting cardinals. In our view, exacting cardinals are just one of a number of hypotheses, conceivably compatible with AC, that refute the

HOD conjecture.

This includes the existence of an elementary embedding from HOD to HOD, which the second author showed in [21] implies the failure of the HOD conjecture in the presence of a strongly compact cardinal. This hypothesis is consistent with AC assuming the consistency of a Reinhardt cardinal with a proper class of supercompact cardinals. One appealing aspect of this result is its analogy with Kunen's theorem that if there is an elementary embedding from L to L , then the L -hypothesis fails.

The existence of an elementary embedding from HOD to HOD can be thought of as the relativization to HOD of the choiceless large cardinal axiom asserting the existence of a Reinhardt cardinal. One can similarly define HOD-analogs of various choiceless large cardinals: for example, HOD-Berkeley and HOD-super-Reinhardt cardinals, and so on. This gives a hierarchy of principles that go beyond the HOD conjecture, though the structure theory of these hypotheses alone does not seem to be as interesting as that of choiceless large cardinals. On the other hand, work of the first author with Sargsyan has revealed that some of these relativized choiceless large cardinals are an aspect of the massive structure theory of the Axiom of Determinacy, hinting at the possibility that choiceless large cardinals themselves might somehow be unified with determinacy.

Finally, there are analogs of I_0 for HOD, things like the existence of an elementary embedding from $\text{HOD}(V_{\lambda+1})$ to itself with critical point below λ . Such hypotheses were considered by Koellner and Woodin [45, Theorem 199]. Ultraexacting cardinals fit into this framework since they can be characterized as follows:

Proposition 5.4. *A cardinal λ is ultraexacting if and only if for all ordinal definable $A \subseteq V_{\lambda+1}$, there is an elementary embedding from $(V_{\lambda+1}, A)$ to itself.*

One can use an elementary embedding from $\text{HOD}(V_{\lambda+1})$, or even just an ultraexacting cardinal, to lift some of the softer consequences of I_0 for $L(V_{\lambda+1})$ to $\text{HOD}(V_{\lambda+1})$. For example, if there is an ultraexacting cardinal, then $\text{HOD}(V_{\lambda+1})$ satisfies the coding lemma and therefore $\Theta^{\text{HOD}(V_{\lambda+1})}$ is a strongly inaccessible cardinal and a limit of measurable cardinals, and more. However, to date there is nothing differentiating the structure theory of $\text{HOD}(V_{\lambda+1})$ from that of $L(V_{\lambda+1})$, so there seems to be no distinctive evidence for the consistency of these hypotheses.

Moreover, to argue against the HOD conjecture, one needs to argue not just for the consistency not only of an ultraexacting cardinal (which follows from I_0),

but also for an ultraexacting cardinal with an extendible cardinal below it.¹⁰ [3] argues that since extendible cardinals and ultraexacting cardinals are each consistent relative to I_0 , an extendible cardinal below an ultraexacting cardinal should be consistent. But this argument relies on intuitions about large cardinals that do not hold for ultraexacting cardinals. The consistency of ultraexacting cardinals from I_0 is proved by restricting the universe of sets so that $\text{HOD}(V_{\lambda+1}) = L(V_{\lambda+1})$. Then by Proposition 5.4 and I_0 , it is easy to see that λ is ultraexacting. But clearly in such a universe there cannot be an extendible cardinal below λ (or any significant large cardinals above λ).

Here there seems to be a clear analogy to $\text{OD}_{\mathbb{R}}$ -determinacy, the principle asserting the determinacy of any game on ω whose payoff is ordinal definable from real parameters. If $I_0(\lambda)$ is the λ -analog of $\text{AD}^{L(\mathbb{R})}$, then by Proposition 5.4, it is natural to think of ultraexactingness as the λ -analog of $\text{OD}_{\mathbb{R}}$ -determinacy. $\text{ZFC} + \text{OD}_{\mathbb{R}}$ -determinacy is consistent relative to $\text{ZFC} + \text{AD}^{L(\mathbb{R})}$, and the proof is just like the proof that an ultraexacting cardinal is consistent relative to I_0 in that it requires passing to a model such that $\text{HOD}(\mathbb{R}) = L(\mathbb{R})$. But we do not take this to mean that $\text{OD}_{\mathbb{R}}$ -determinacy is consistent with extendible cardinals, because extendible cardinals imply the existence of complicated and poorly understood sets of reals. Similarly, they imply the existence of complicated subsets of $V_{\lambda+1}$, and therefore we should not jump to the conclusion that ultraexacting cardinals are consistent with extendible cardinals (above or below).

Therefore this potential RNA of the HOD conjecture requires something more. [3] proposes structural reflection as a way of showing that ultraexacting cardinals are natural large cardinal hypotheses, and therefore presumably just as acceptable as, for example, supercompact cardinals. We discuss the structural reflection program in Section 6.1.

Woodin's Theorem 4.1 is like Scott's theorem prior to the certification of the consistency of measurable cardinals. Claiming that it is a definitive refutation gives rise to a circularity concern. By Aguilera-Bagaria-Lücke, we know its hypothesis is consistent relative to choiceless cardinals which the HOD conjecture implies are inconsistent. What's missing is something to play the role of $0^\#$ or $L[U]$ either for choiceless cardinals or for ultraexacting cardinals.

Aguilera-Bagaria-Lücke conjecture that an ultraexacting cardinal above an extendible cardinal is at least as strong consistency-wise as a rank-Berkeley cardinal.

¹⁰In fact, a strongly compact cardinal suffices here.

If that's right, whatever plays the role of $L[U]$ would seem to be a long way off.

So the analogue of $L[U]$ is missing. Another difference from the L Hypothesis case is that no exact structural characterization or reversals are known for the hypothesis of Theorem 4.1. Equivalences and reversals would make ultraexacting cardinals more compelling. An essential question right now is whether the hypotheses of Theorems 4.1 and 3.1 are in any sense optimal:

Question 5.2. Does the existence of traditional large cardinals plus the failure of the HOD Hypothesis imply the consistency of ultraexacting cardinals?

It must be stressed that the lack of equivalences and reversals is a historical contingency. Ultraexacting cardinals are too new to expect a detailed theory of them, and they are so powerful, in terms of consistency strength, that substantial mathematics needs to be developed to even formulate interesting statements they might be equivalent to. A theorem's taking time to be recognized as what it purports to be is not unheard of in ordinary mathematical practice. Foreshadowing later discussion, the Clay Millennium Prize rules acknowledge that assessing the correctness of a solution to an important open problem and seeing how it fits into broader mathematics—in short, the *acceptance* of a solution—takes time. As markers of meeting the threshold for acceptance, the rules state that publication in a peer-reviewed journal, citations from other mathematicians building on the notions introduced in giving the solution, and conferences held to disseminate the new notions and techniques are expected.

The missing pieces—the inner model, the equivalences, and the reversals—leave the HOD conjecture with an open character. If a proof of the HOD conjecture was found tomorrow, it would be no more surprising after the results of [3] than it would have been before. That is not to say that choiceless cardinals being proved inconsistent would be unsurprising, but that is part of why the HOD conjecture is such a strong claim. Moreover, if the HOD conjecture were refuted assuming a standard large cardinal hypothesis, *that* would be the generally accepted refutation.

By contrast, a proof of the L conjecture would be calamitous. It would mean the theory of $L[U]$ is vacuous despite all the information we have suggesting it is not.

In [3], exacting and ultraexacting cardinals are presented as on one side of a new dividing line in the large cardinal hierarchy between those large cardinals which imply $V \neq \text{HOD}$ and those that are consistent with $V = \text{HOD}$. This is in analogy

with large cardinal hypotheses which imply $V \neq L$ and those which can hold in L . The authors of [3] should agree that what was needed to certify measurables and legitimize the earlier dividing line is needed to certify an exacting cardinal above an extendible cardinal.

6 What is a genuine large cardinal hypothesis?

There is no formal definition of *large cardinal hypothesis*.¹¹ It seems reasonable to think that there cannot be a precise syntactic criterion for being a large cardinal hypothesis: one would hope that the collection of large cardinal hypotheses is cofinal in the consistency strength hierarchy, and therefore the set of large cardinal hypotheses is not recursively enumerable. Set theorists have learned to live with treating large cardinals as unified by family resemblance rather than necessary and sufficient conditions.

What seems to be important is that the hypotheses form a hierarchy, that there is only one way of going up. An aspect of this picture is the joint consistency of all large cardinal hypotheses.

At the furthest reaches of the presently known large cardinal hierarchy lie choiceless cardinals and generalizations of I_0 . The traditional picture is that the choiceless cardinals, if they are consistent at all, transcend all large cardinal hypotheses compatible with ZFC, but in fact, it is unknown whether there are generalizations of I_0 that reach the level of a Reinhardt cardinal, for example.

Ultraexacting cardinals do not fit neatly into either the choiceless hierarchy or the I_0 hierarchy. As of now, there is no ultraexacting hierarchy to speak of, just

¹¹Woodin gave the following operational definition in [44].

Definition 6.1 ([44, Definition 100]). $(\exists x\varphi)$ is a *large cardinal axiom* if

1. $\varphi(x)$ is a Σ_2 -formula;
2. (As a theorem of ZFC) if κ is a cardinal such that

$$V \models \varphi[\kappa]$$

then κ is strongly inaccessible and for all partial orders $\mathbb{P} \in V_\kappa$,

$$V^\mathbb{P} \models \varphi[\kappa].$$

This has caused some confusion since this is just a necessary condition; it is not a sufficient criterion for $\exists x\varphi$ to be a large cardinal axiom (nor was it Woodin's intention to give one). For example, under Woodin's definition, the existence of an inaccessible cardinal below which the GCH fails on a tail counts as a large cardinal axiom.

the two hypotheses (exacting and ultraexacting) isolated by Aguilera, Bagaria, and Lucke. Do these hypotheses open a third path up the consistency strength hierarchy, or are they isolated phenomena that hold only in the specific models constructed in [3]? Aguilera, Bagaria, and Lücke argue that ultraexacting and exacting cardinals are indeed “natural large cardinals” [3, Abstract] by casting them as structural reflection principles. We next assess these arguments.

6.1 Structural Reflection Principles

Aguilera-Bagaria-Lücke aim to certify that (ultra)exacting cardinals are genuine large cardinal hypotheses by 1) proving equivalents, 2) proving theorems that place them in the system of choiceless cardinals, and 3) deriving them from *Structural Reflection* principles.

Bagaria’s structural reflection program has “the ultimate objective of showing that all large cardinals are in fact different manifestations of a single general reflection principle” [5, p. 30]. Gödel gestured at the possibility of such a principle:

It is certainly impossible to give a combinational and decidable characterization of what an axiom of infinity is; but there might exist, e.g., a characterization of the following sort: An axiom of infinity is a proposition which has a certain (decidable) formal structure and which in addition is true. Such a concept of demonstrability might have the required closure property, i.e., the following could be true: Any proof for a set-theoretic theorem in the next higher system above set theory (i.e., any proof involving the concept of truth which I just used) is replaceable by a proof from such an axiom of infinity. It is not impossible that for such a concept of demonstrability some completeness theorem would hold which would say that every proposition expressible in set theory is decidable from the present axioms plus some true assertion about the largeness of the universe of all sets. [17]

A candidate for the formal structure of large cardinal hypotheses is *reflection*. Large cardinals have long been informally motivated as “reflection principles,” and Gödel thought that reflection provided a unified way of encapsulating large cardinals, saying to Wang that “in the last analysis, every axiom of infinity should be derivable from the (extremely plausible) principle that V is undefinable” [40, p. 285]. Tait took this as a mandate and started deriving large cardinal hypotheses

from syntactic reflection principles, the idea being that the reflection principles lend intrinsic justification to the large cardinals derived from them [37, 38, 39]. By varying parameters in the reflection principle template, one could derive stronger large cardinals. Koellner established severe limitations on the program: The syntactic reflection principles cannot justify large cardinals incompatible with $V = L$ before they become outright inconsistent [27].

Bagaria introduced a form of reflection having to do with the existence of mathematical structures in V . The basic template is as follows.

Definition 6.2 (Structural Reflection Template). Let C be a class of structures of the same signature. Then there is a rank κ such that for every structure $A \in C$, there is a $B \in C \cap V_\kappa$ and an elementary embedding $j : B \rightarrow A$.

By tweaking this principle in various ways, one gets structural reflection principles which reformulate large cardinals. The structural reflection program has succeeded in reformulating large cardinal hypotheses throughout the hierarchy.¹²

There are some difficulties with the justificatory role the structural reflection program is intended to play in [3]. First, it is not unanimously agreed that large cardinals require intrinsic justification, or that there is such a thing as intrinsic justification of set theoretic axioms at all. Indeed, for many large cardinal set theorists, large cardinal hypotheses are justified by their consequences and the intricate and fundamental theory of consistency strength built upon them, not by their formal structure.

Second, long predating the structural reflection program, there have been attempts to uniformly express large cardinal hypotheses, programs which moreover did not face the limitations of Tait’s program. For example, in the 1970s, Abramson, Harrington, Kleinberg, and Zwicker reformulated measure and partition large cardinal notions incompatible with $V = L$ in terms of combinatorial “flipping” properties [2]. [15] even “flipped” supercompact cardinals. The flipping program did not increase the community’s confidence in large cardinals, and the language of flipping was not used to specify new hypotheses.

¹²Philip Welch and Leon Horsten have used their own *Global Reflection Principle*, a modification of the notion of *subcompact cardinal*, to (purportedly) intrinsically justify large cardinals incompatible with $V = L$ [42]. Welch shows, with the goal of intrinsically justifying axioms of definable determinacy, that their principle implies the existence of unboundedly many measurable Woodin cardinals [41, 43].

Another attempt at uniformization led to the formulations whose purported un-intuitiveness motivate the structural reflection program: elementary embeddings of the universe into class inner models. This unifying framework was developed by Solovay and Reinhardt and expounded in [36]. It has been extremely successful and is the part of the motivation for most of the strong large cardinal hypotheses used today, including strong cardinals, Woodin cardinals, and supercompact cardinals. One potential issue with this approach is its ascent into inconsistency: one of the most natural principles from the perspective of this framework is a Reinhardt cardinal, the very hypothesis refuted by Kunen’s inconsistency theorem.

Third, as Hou has observed [24], the structural reflection program has not yet achieved Gödel’s dream of a single template from which the known large cardinals are derivable. The collection of structural reflection principles displays the non-uniformity of the large cardinal hypotheses. A given structural reflection principle allows the derivation of principles up to some point in the large cardinal hierarchy but no more. Then a new structural reflection principle has to be formulated to derive stronger principles. For example, the *Exact Structural Reflection Principle* is used to derive exacting cardinals, but the *Square Root Exact Structural Reflection Principle* is needed to yield ultraexacting cardinals.

The most fundamental worry about the role of structural reflection in [3] has to do with consistency. Deriving (ultra)exacting cardinals from Square Root Exacting Structural Reflection is intended to make the principles seem natural and true. Hence we need not worry about their consistency and can ignore the tacit conjecture. But Jensen’s quote above about the Kunen Inconsistency is a warning: A principle that seems natural and true can nonetheless be inconsistent. *Derivations from structural reflection principles do not obviate the need for certifications of consistency.*¹³

It is not hard to devise structural reflection principles that are inconsistent with the Axiom of Choice. In this sense, the structural reflection program does not display any advantage over the Reinhardt–Solovay elementary embedding approach. For example:

Definition 6.3. The *rank Berkeley structural reflection axiom* holds at λ , denoted

¹³On this point, there is no tension between the “innocent until proven inconsistent” attitude of large cardinal set theorists and the demand for consistency here. Choiceless cardinals *have* been proved inconsistent! The question at issue is whether they are consistent in ZF. But the Kunen Inconsistency and the even easier refutation of Berkeley cardinals in ZFC are compelling reasons for caution.

$\text{RBSR}(\lambda)$, if there is some $\mu < \lambda$ such that for any class $\mathcal{C}$ that is Σ_2 -definable from parameters in $V_\mu \cup \{\lambda\}$, for all structures $\mathcal{N} = (N, B, \dots) \in \mathcal{C}$ in a finite signature with $|B| < \lambda$, for some $\mathcal{M} = (M, A, \dots) \in \mathcal{C}$ with $|A| < \mu$, there is an elementary embedding from $\mathcal{M}$ to $\mathcal{N}$.

Then $\text{RBSR}(\lambda)$ immediately contradicts the Kunen inconsistency theorem, and in fact the least λ such that $\text{RBSR}(\lambda)$ holds is exactly the least rank Berkeley cardinal. This issue is also recognized by Aguilera–Bagaria–Lücke, who formulate a structural reflection principle characterizing the least Berkeley cardinal [3, Proposition 5.10].

A somewhat subtler plausibility argument for the consistency of ultra-exacting cardinals above extendible cardinals is given in [3, Section 1.2]. It runs as follows. Extendible cardinals are consistent relative to I_2 . Exacting cardinals are also consistent relative to I_2 . Moreover, both extendible cardinals and exacting cardinals are natural large cardinal hypotheses (as verified by their structural reflection characterization), so since each is consistent on its own, the two should be jointly consistent.

Similar plausibility arguments can be used to justify inconsistent hypotheses.

Definition 6.1. A cardinal λ is *exacting relative to a set Y* if there is some $\mu < \lambda$ such that for any family $\mathcal{C}$ of structures Σ_2 -definable over V from parameters in $V_\mu \cup \{Y\}$, for any $N \in \mathcal{C}$ of type (λ, δ) with $\delta < \lambda$, for some $M \in \mathcal{C}$ of type $(\lambda, \bar{\delta})$ with $\bar{\delta} < \mu$, there is an elementary embedding from M into N .¹⁴

An argument due to Aguilera shows that λ is exacting if and only if λ is exacting relative to $\emptyset$. In our view, this is the best-motivated characterization of exacting cardinals in terms of structural reflection.

Setting structural reflection aside, we have the following equivalences:

Proposition 6.2. *If λ is a cardinal and Y is a set, the following are equivalent:*

- (1) λ is exacting relative to Y .
- (2) For any nonempty set $A \subseteq V_{\lambda+1}$ that is ordinal definable from Y , there exist $x, y \in A$ and a nontrivial elementary embedding from (V_λ, x) to (V_λ, y) .
- (3) For every ordinal $\alpha > \text{rank}(Y)$, for some $X \preceq V_\alpha$ with $V_\lambda \cup \{\lambda, Y\} \subseteq X$, there is an elementary $j : X \rightarrow V_\alpha$ with $\text{crit}(j) < \lambda$, $j(\lambda) = \lambda$ and $j(Y) = Y$.

¹⁴A structure $(M, A, \dots)$ has type (λ, δ) if $|M| = \lambda$ and $|A| = \delta$.

Relative to which sets Y can a cardinal λ be exacting? In ZF, if λ is club Berkeley, then λ is exacting relative to every set Y , while in ZFC, every exacting cardinal has countable cofinality, and it is impossible for λ to be exacting relative to a cofinal ω -sequence in λ . So one might argue that principles stating that λ is exacting relative to “many” sets Y are ZFC-approximations to club Berkeley cardinals, in analogy to Aguilera, Bagaria, and Lucke’s claim that exacting cardinals are ZFC-approximations to rank Berkeley cardinals.

Definition 6.3. If λ is a cardinal and $\gamma > \lambda$ is regular, then λ is γ -stationarily exacting if for all ordinals α , λ is exacting relative to stationarily many $Y \in P_\gamma(V_\alpha)$.

Extending an argument due to Aguilera, Bagaria, and Lücke, one can prove that if $\text{ZFC} + I_2$ is consistent, then so is ZFC plus the existence of a stationarily exacting cardinal:

Proposition 6.4. *If $j : V \rightarrow M$ is an I_2 -embedding with critical point κ , then there is a Prikry generic $g \subseteq \kappa$ such that for all regular $\gamma \geq \kappa^+$, κ is γ -stationarily exacting in $V[g]$.*

Proof. Let $j_{0\omega} : V \rightarrow M_\omega$ be the ω -th iterate of j and let $g = \{\kappa, j(\kappa), j(j(\kappa)), \dots\}$ be the critical sequence of j . Then over M_ω , g is a Prikry generic subset of $\lambda = j_{0\omega}(\kappa)$. Therefore it suffices to show that for all regular $\gamma \geq \lambda^+$, λ is γ -stationarily exacting in $M_\omega[g]$.

Working in $M_\omega[g]$, we first claim that for any α , club many $Y \in P_\gamma(M_\omega[g]_\alpha)$ are Σ_2 -definable from parameters in M_ω . It suffices to prove this for $\alpha > \gamma$. First, note that the set of $Y \preceq M_\omega[g]_\alpha$ such that $\lambda \cup \{\mathbb{P}\} \in Y$ contains a club C in $M_\omega[g]$. Therefore $D = \{Y \cap M_\omega : Y \in C\}$ is club in $P_\gamma((M_\omega)_\alpha)$, and since $M_\omega[g]$ is a γ -cc forcing extension of M_ω , D contains a club $E \in M_\omega$. For any $X \in E$, note that $X[g]$ is the unique $Y \preceq M_\omega[g]_\alpha$ with $Y \cap M_\omega = X$, and so $X[g]$ is Σ_2 -definable in $M_\omega[g]$ from parameters in M_ω . Finally, the previous sentence also implies that the set $F = \{X[g] : X \in E\}$ is equal to $\{Y \preceq M_\omega[g]_\alpha : Y \cap (M_\omega)_\alpha \in E\}$, and hence F is club in $P_\gamma(M_\omega[g]_\alpha)$ and every $Y \in F$ is Σ_2 -definable from parameters in M_ω , proving the claim.

We will verify that for any $Y \in M_\omega$, λ is exacting relative to Y in $M_\omega[g]$. Combined with the previous paragraph, this yields that λ is γ -stationarily exacting in $M_\omega[g]$. (In fact, λ is “ γ -club exacting” in the natural sense.)

Fix $n < \omega$ large enough that $j_n(Y) = Y$. We claim that $\mu = \text{crit}(j_n)$ witnesses that λ is exacting relative to Y in $M_\omega[g]$. Suppose $\mathcal{C}$ is a set of structures definable

in $M_\omega[g]$ from parameters in $V_\mu \cup \{Y\}$. Then $j_n(\mathcal{C}) = \mathcal{C}$. Suppose towards a contradiction that $\delta < \lambda$ is least such that for some $N \in \mathcal{C}$ of type (λ, δ) , there exist no $\bar{\delta} < \mu$ and $M \in \mathcal{C}$ of type $(\lambda, \bar{\delta})$ such that M elementarily embeds into N . Since δ is definable in M_ω from $\mathcal{C}, \mu$, and λ , which are all in the range of $j_{n+1, \omega}$, it follows that $\delta < \text{crit}(j_{n+1, \omega}) = j_n(\mu)$. Let $N' = j_n(N)$. Then $N' \in \mathcal{C}$. Moreover, j_n restricts to an elementary embedding from N into N' .

We would like to argue that $M_\omega[g]$ satisfies that there is an elementary embedding from N to N' ; the issue is that $j_n \upharpoonright N \notin M_\omega[g]$. But by a theorem of Woodin [29, Corollary 1.5], $M_\omega[g]$ is Σ_2^1 -correct at λ , and therefore $M_\omega[g]$ satisfies that there is an elementary embedding from N to N' . Since $N' = j_n(N)$ and $M_\omega[g]$ satisfies that for some $\delta < j_n(\mu)$, there is a structure of type (λ, δ) in $\mathcal{C}$ that elementarily embeds into N' , $M_\omega[g]$ satisfies that for some $\bar{\delta} < \mu$, there is a structure of type $(\lambda, \bar{\delta})$ in $\mathcal{C}$ that elementarily embeds into N . $\square$

Does the structural reflection characterization of stationarily exacting cardinals combined with the consistency proof Proposition 6.4 make stationary exactingness a natural large cardinal hypothesis? Perhaps not, but it is unclear why the Aguilera–Bagaria–Lucke arguments in favor of exacting and ultra-exacting cardinals are preferable.

In any case, we have the following inconsistency theorem:

Proposition 6.5. *Suppose κ is extendible, $\lambda > \kappa$ is a cardinal, and $\gamma > \lambda$ is regular. Then λ is not γ -stationarily exacting.*

Proof. Assume towards a contradiction that λ is γ -stationarily exacting. In particular, there is an elementary embedding from V_λ to V_λ with critical point less than λ . As a consequence of the Kunen inconsistency theorem, we then have that λ is a singular strong limit cardinal.

We claim that if U is a γ -complete ultrafilter on an ordinal ν , then for a club of $Y \in P_\gamma(V_{\nu+2})$, U is ordinal definable from Y . To see this, suppose $Y \in P_\gamma(V_{\nu+2})$ is such that $U \in Y$ and for any distinct ultrafilters W_0 and W_1 that belong to Y , $W_0 \cap Y \neq W_1 \cap Y$; there is a club of such Y . Letting $A = \bigcap_{B \in U \cap Y} B$ and $\alpha = \min A$, U is the unique ultrafilter $W \in Y$ such that α belongs to every $B \in W \cap Y$; this shows that U is ordinal definable from Y .

For any cardinal ρ , let $\text{CD}(\rho)$ denote the class of sets that are definable from ρ -complete ultrafilters on ordinals. Let $\text{CD} = \bigcap_\rho \text{CD}(\rho)$ and let HCD be the class of hereditarily CD sets. By Proposition 4.11 and Theorem 4.15 of [22] combined

with the extendibility of κ , V is a forcing extension of HCD via a forcing of size less than λ . In particular, λ is singular in HCD, and so there is a cofinal set $C \subseteq \lambda$ with $\text{ot}(C) < \lambda$ such that C is definable from a γ -complete ultrafilter U on an ordinal ν .

Since λ is γ -stationarily exacting and U is ordinal definable from club many $Y \in P_\gamma(V_{\nu+2})$, there is some $Y \in P_\gamma(V_{\nu+2})$ such that λ is exacting relative to Y and U is ordinal definable from Y .

Since λ is exacting relative to Y , there is a nontrivial elementary embedding $j : (V_\lambda, C) \rightarrow (V_\lambda, C)$. By iterating j , we can ensure that $\text{crit}(j) > \text{ot}(C)$. Then every element of C is fixed by j . But by the Kunen inconsistency theorem, j has no fixed points between $\text{crit}(j)$ and λ . This contradicts that C is cofinal in λ . $\square$

We are able to refute the existence of a stationarily exacting cardinals above an extendible because the analog of the HOD conjecture for the inner model HCD has already been proved. If the HOD conjecture itself is proved, the refutation of exacting cardinals above extendibles will proceed in exactly the same way.

Aguilera, Bagaria, and Lucke's case for exacting cardinals has three components: the argument via structural reflection that exacting cardinals are natural large cardinal hypotheses; the proof that exacting cardinals are consistent relative to accepted hypotheses like I_2 ; and the proof that granting the consistency of large cardinals beyond choice, exacting cardinals are consistent with essentially all large cardinals. The example of stationarily exacting cardinals casts doubt on the idea that the first two components alone secure the consistency of exacting cardinals above extendible cardinals: these two components extend in a seemingly identical way to stationarily exacting cardinals, yet the existence of a stationarily exacting cardinal above an extendible cardinal is inconsistent. But for the third component to secure the consistency of exacting cardinals above extendible cardinals, one must take for granted that large cardinals beyond choice are consistent, and hence that the HOD conjecture is false.

7 Justifying large cardinals in light of the Kunen inconsistency

Was Woodin wrong to think he had not refuted the HOD conjecture? The answer depends on what he conceived a refutation to be and what evidence was at hand for believing in the consistency of the hypothesis of Theorem 4.1.

In 2004, Woodin laid out precise stipulations for collecting a cash prize for a solution to an early version of the HOD conjecture:

Question 32 (Woodin) Suppose that there is an extendible cardinal. Must HOD compute the successor correctly for some (uncountable) cardinal?

Prize:

$$\$1000[\max(\min(n, 10 - n), 1)]$$

where $n = (\text{calendar year of submission}) - 2004$.

Terms: Collect if a correct proof is given for either “yes”, or if a correct proof is given that the failure implies the consistency with ZFC of the large cardinal I_0 of Kanamori’s book. (Details: Clay rules) [16, p. 320]

A refutation of the HOD conjecture is a relative consistency proof showing that if $\text{ZFC} + \text{“there is an extendible cardinal”} + \neg\text{HOD Hypothesis}$ is consistent, so is $\text{ZFC} + \exists \lambda I_0(\lambda)$. As of writing, nothing is known about this consistency strength lower bound.

Question 7.1. Assume ZFC. Does the existence of a supercompact cardinal plus the failure of the HOD Hypothesis have large cardinal strength greater than a supercompact cardinal? Does it imply there is an inner model of NBG + “there is a Reinhardt cardinal”?

Woodin’s position is counterintuitive. Normally higher consistency strength entails lower credence in consistency. Thus Woodin’s original proof of Projective Determinacy was not seen as closing the question of whether Projective Determinacy is consistent. (At the time it was expected to be stronger than the known ZFC large cardinals.) The proof from I_0 is notable more for the isolation of I_0 than for boosting set theorists’ confidence in Projective Determinacy; Martin and Steel are credited with “the” proof of Projective Determinacy in no small part because of its optimality [32]. The Martin-Steel theorem brought the hypothesis down from one approaching the Kunen Inconsistency to one for which there was inner model theory.

Generally, if the negation of a statement has consistency strength at least that of a theory approaching the Kunen Inconsistency, that fact is naturally interpretable as evidence that the negation of the statement is *inconsistent*. But Woodin interpreted it as evidence for its consistency.

If there were such a consistency proof, the thought goes, then the proof must reveal some structure giving reason to believe that $\neg$ HOD Hypothesis is consistent, despite its surpassing—on the knowledge of the time—the ZFC large cardinal hierarchy. It would have to simultaneously give reason to believe in the consistency of choiceless hypotheses. This is in line with the notion of RNA above. (According to the prize conditions, Woodin would not be obligated to pay for what we have called “perhaps the most definitive conceivable refutation” above!)

An RNA as Woodin envisaged would conform with the Clay rules, a relevant part of which, as they were stated in 2004, enforce that a refutation thoroughly refute the *idea* behind a problem rather than provide a counterexample which is easily “monster-barred.”

In the case that a mathematician discovers a proposed counterexample to a prize problem, the [Scientific Advisory Board] will consider this counterexample after publication and the same two-year waiting period as for a proposed solution will apply. If the counterexample shows that the original problem survives after reformulation or elimination of some special case, then the SAB may recommend that a small prize be awarded to the author. [1]

As it pertains to the HOD conjecture, the Clay rules require that stronger ZFC large cardinals cannot be invoked to monster-bar the refutation. Woodin’s notion of refutation meets this requirement. For a refutation in his sense will substantiate the consistency of the choiceless context it gives rise to. Combined with Woodin’s theorems in §2.2, the consistency then refutes the provability of the HOD Hypothesis in ZFC plus large cardinals.

All that being said, there still seems to be a tension between Woodin’s acceptance of axiom I_0 , and the role it plays in his conception of what a refutation is, and his refusal to consider $W(\lambda)$ (i.e., the existence of ultraexacting cardinals) a genuine large cardinal hypothesis. What meaningfully differentiates these?

Woodin’s approach to generalizing large cardinal hypotheses below the Kunen Inconsistency and above axiom I_1 has been to build a hierarchy of principles approximating a $j : V_{\lambda+2} \rightarrow V_{\lambda+2}$. The hierarchy grows according to the hierarchy of definability. Since there cannot exist a $j : V_{\lambda+2} \rightarrow V_{\lambda+2}$, Woodin’s approach to formulating principles consistent with choice was to restrict to the constructible fragment of $V_{\lambda+2}$. Thus the starting point was $L(V_{\lambda+1})$. Following the predictive

analogy with determinacy theory, enlargements of $L(V_{\lambda+1})$ are obtained by constructing from $V_{\lambda+1}$ and terms for sets $X \in V_{\lambda+2}$ which are universally Baire after collapsing λ to be countable.

More recently, Woodin has defined “internal” generalizations of I_0 which challenge the current approach to inner model theory. It is not known if they transfer from V to Ultimate L (assuming the Ultimate L conjecture), although they can hold in a set forcing extension of Ultimate L . One generalization, axiom $(I_0^\infty)^+$, is motivated by the predictive analogy with AD and by Woodin’s *Sealing Theorem*, and it is quite technical to state.¹⁵ Conditioned on further conjectures, the Ultimate L conjecture refutes it [47, Theorem 5.7]. Yet Woodin is committed to its being a genuine large cardinal hypothesis, modulo consistency. He conjectures its consistency is tantamount to the Ultimate L conjecture failing¹⁶ and writes

... the Axiom $(I_0^\infty)^+$ is a candidate for an arguably natural large cardinal axiom in the context of the Axiom of Choice, and much more interestingly even in the context of $V = \text{HOD}$, which is beyond the reach of the Inner Model Program. [47]

The qualification “even in the context of $V = \text{HOD}$ ” attests to his awareness and dismissal of exacting cardinals prior to their rediscovery by Aguilera-Bagaria-Lücke.

The apotheosis of the I_0 hierarchy—the point to which the principles lead—is the Kunen Inconsistency. This has given the sense, which Woodin’s envisioned refutation draws on, that ZFC plus the rank-to-rank hierarchy represents the furthest reaches of the ZFC large cardinal hierarchy. By contrast, $W(\lambda)$ lacks the context in which I_0 was defined and which justifies it. It “jumps ahead.” Why should Woodin have believed it is consistent, much less true?

The crux of the issue of whether either Theorem 3.1 or 4.1 is an RNA of the HOD conjecture is whether an exacting cardinal or $W(\lambda)$ is consistent above a HOD-supercompact cardinal. Theorem 3.2 pushes this question back to the consistency of choiceless cardinals. Is it even possible to justify choiceless cardinals in light of the Kunen Inconsistency?

¹⁵Unwind [47, Definition 5.3].

¹⁶The joint consistency of $(I_0^\infty)^+$ with other large cardinals is not taken for granted: “Implicit in this conjecture is the conjecture the Axiom $(I_0^\infty)^+$ holding at λ is not refuted by any additional large cardinal hypothesis combined with $V_\lambda \prec V$.”

7.1 Towards consistency

Since [45] was published, the most significant advance in the upper reaches of large cardinals is Schlutzenberg’s theorem.

Theorem 7.2 (Schlutzenberg [34]). $\text{ZF} + \exists j : V_{\lambda+2} \rightarrow V_{\lambda+2}$ is consistent if and only if $\text{ZFC} + \exists \lambda I_0(\lambda)$ is consistent.

To prove the theorem, Schlutzenberg constructed a canonical model with a $j : V_{\lambda+2} \rightarrow V_{\lambda+2}$. The proof is an ingenious adaptation of directed systems arguments in the setting of very large cardinals.

Apart from its intrinsic interest, Schlutzenberg’s theorem changes the picture of how choiceless cardinals might relate to ZFC large cardinals. It shows that going past the Kunen Inconsistency does not necessarily mean working above the hierarchy of ZFC large cardinals; Choiceless cardinals might not represent consistency strengths which cannot be accessed assuming the axiom of choice.

Question 7.3. Are ZFC large cardinals and choiceless cardinals interwoven cofinally in the consistency strength hierarchy?¹⁷

Recall from §2.2 that the HOD conjecture does not imply that a Reinhardt cardinal cannot exist, so Schlutzenberg’s theorem does not refute the HOD conjecture. But it opens the door to definitively refuting it. Woodin, inspired by Schlutzenberg’s theorem, has speculated a way of building more canonical models for stronger Reinhardt embeddings using a hierarchy of enlargements of $L(V_{\lambda+1})$ [47]. The hope is to ultimately construct a model with a Reinhardt cardinal by folding in enough rank-to-rank embeddings.

These models would be canonical in essentially the familiar loose sense from inner model theory. They should admit a fine structural analysis akin to Steel’s analysis of $L(\mathbb{R})$. Then the consistency of choiceless cardinals would not imply an end to inner model theory, despite what [6] asserts. The program of studying consistency strengths via inner models will not be shown to be impossible if e.g. it is consistent for there to be a Reinhardt cardinal above an extendible. The Ultimate L conjecture will, on the other hand, be false.

The Kunen Inconsistency would then say that elementary embeddings are eventually a restrictive language with which to formulate large cardinals in ZFC.

¹⁷[6] speculates that perhaps the HOD-versions of choiceless cardinals specifically are cofinal.

The authors, in a different direction, speculate that choiceless cardinals exist in large models of AD.

The first author and Grigor Sargsyan have shown that many cardinals have Berkeley properties assuming AD [8]. Of particular interest is that in the standard models of AD, ω_1 is Θ -Berkeley, or Berkeley with respect to structures of height less than Θ . In the Nairian models of [9], it may be that ω_1 is Berkeley past Θ and through all the ordinals.

Conjecture 7.4. ω_1 is an Ord-Berkeley cardinal in the full Nairian model.

Investigating choiceless cardinal phenomena in models of determinacy seems a reasonable way to gather evidence for their consistency. The ultimate goal of this line of research would be to unify the theories of determinacy and large cardinals beyond choice.

Another class of evidence comes from investigating the theories of choiceless cardinals directly. The second author and Farmer Schlutzenberg showed that if there is a Reinhardt cardinal, then the powerset operation is eventually periodic with period 2 [23]. This means that sufficiently large consecutive levels of the cumulative hierarchy have markedly different structure. (It is in this paper that rank-Berkeley cardinals were first isolated.) The second author, building on work of Woodin [45] and Cutolo [12, 13, 14], has shown that if there is a Reinhardt cardinal, then there are arbitrarily large measurable successor cardinals [20] and the supremum of the well-founded relations on a rank V_α is a strong limit cardinal if and only if α is odd, for all sufficiently large ordinals α [18].

7.2 On the failure of inner model theory

If choiceless cardinals are consistent, must we give up “the prospect of ever having detailed, fine-structural insight into V ”?

If inner model theory is the project of showing that every genuine large cardinal axiom holds in the HOD of a determinacy model—which is Woodin’s explication of it—and if choiceless cardinals are genuine large cardinals, then the consistency of choiceless cardinals of course implies that project cannot succeed. There cannot be a Reinhardt cardinal in the HOD of a determinacy model.

Woodin’s explication of the Inner Model Program has the advantage of being precise enough that one day it may be shown impossible, but its precision means that a refutation need not satisfy the Clay rules. The Ultimate L conjecture being false

would not rule out the possibility that there exist canonical models with supercompact cardinals satisfying the Ultrapower Axiom¹⁸ and which are *not* weak extender models. Indeed, it is possible that there are inner models for supercompacts which are not weak extender models but nonetheless satisfy the axiom $V = \text{Ultimate } L$. Although the axiom $V = \text{Ultimate } L$ is incompatible with Reinhardt cardinals, there may be ZFC large cardinals at the same consistency strength as Reinhardt cardinals. This possibility is raised by Schlutzenberg’s theorem. So the consistency of choiceless cardinals does not preclude inner models satisfying $V = \text{Ultimate } L$ from going all the way up the consistency strength hierarchy.

These models would not be universal in the sense that weak extender models are. They would not absorb all extenders in the ambient universe. And they would not represent the completion of the inner model program. They could nonetheless serve for supercompact cardinals the same function $L[U]$ serves for measurables. The key point is that without knowing how to build the inner model for a supercompact cardinal, one cannot argue that the failure of the Ultimate L conjecture implies the failure of inner model theory.

What we have argued would secure the refutation of the HOD conjecture—the consistency of choiceless cardinals—are canonical models constructed from generalizations of I_0 or from extremely large models of determinacy. In other words, we have argued that inner model theory—more broadly construed than Woodin takes it—is essential to refuting the system of conjectures whose positive solutions would make it difficult to deny that the universe of set theoretic discourse just is a weak extender model.

The “two futures” of [6] are a false dichotomy.

8 Conclusion

Whether choiceless cardinals are consistent is what the HOD conjecture is really about. Aguilera, Bagaria, and Lücke have put the problem into relief by showing which choiceless hypothesis is sufficient to prove the consistency of Woodin’s hypothesis $W(\lambda)$, or equivalently an ultraextending cardinal. But justifying the consistency of the choiceless hypothesis would refute the HOD conjecture directly. Theorem 4.1

¹⁸A combinatorial principle, isolated by the second author, sufficient to develop inner model theory via comparison of structures iterable by ultrapowers. The Ultrapower Axiom is expected to hold in any model containing large cardinals and with a reasonable claim to being canonical. See [19].

would then be a wheel spinning idly.¹⁹ For it depends for its legitimacy on a piece of mathematics which would by itself cleanly refute the HOD conjecture. This is why Woodin did not think of the theorem as refuting the HOD conjecture when he proved it.

The most convincing RNA of the HOD conjecture would be to show that choiceless cardinals are consistent, i.e., to provide strong evidence against the implicit conjecture.²⁰ This will not be done in one fell swoop. It requires giving not a proof but an argument drawing on theorems that choiceless cardinals are consistent. This paper has outlined the beginning of such an argument.

The HOD conjecture is still open.

References

- [1] Rules for the millennium prizes. https://web.archive.org/web/20040606155616/http://www.claymath.org/millennium/Rules_etc/#expand. Accessed: 25 April 2025.
- [2] Fred G Abramson, Leo A Harrington, Eugene M Kleinberg, and William S Zwicker. Flipping properties: a unifying thread in the theory of large cardinals. *Annals of Mathematical Logic*, 12(1), 1977.
- [3] Juan P Aguilera, Joan Bagaria, and Philipp Lücke. Large cardinals, structural reflection, and the hod conjecture. *arXiv preprint arXiv:2411.11568*, 2024.
- [4] Navin Aksornthong, Takehiko Gappo, James Holland, and Grigor Sargsyan. On ω -strongly measurable cardinals in $\mathbb{P}_{\max}$ extensions. *Journal of Mathematical Logic*, page 2450018, 2024.
- [5] Joan Bagaria. Large cardinals as principles of structural reflection. *Bulletin of Symbolic Logic*, 29(1):19–70, 2023.
- [6] Joan Bagaria, Peter Koellner, and W Hugh Woodin. Large cardinals beyond choice. *Bulletin of Symbolic Logic*, 25(3):283–318, 2019.

¹⁹“Here I should like to say: a wheel that can be turned though nothing else moves with it, is not part of the mechanism.” §272, *Philosophical Investigations*.

²⁰The proponent of Structural Principles should agree, given their program’s prediction that Berkeley cardinals are consistent.

- [7] Omer Ben-Neria and Yair Hayut. On ω -strongly measurable cardinals. *Forum of Mathematics, Sigma*, 11:e19, 2023.
- [8] Douglas Blue and Grigor Sargsyan. AD^+ implies ω_1 is club Θ -Berkeley. Forthcoming, 2024.
- [9] Douglas Blue, Paul Larson, and Grigor Sargsyan. Nairian models. Forthcoming.
- [10] Yong Cheng, Sy-David Friedman, and Joel David Hamkins. Large cardinals need not be large in hod. *Annals of Pure and Applied Logic*, 166(11):1186–1198, 2015.
- [11] James Cummings, Sy David Friedman, and Mohammad Golshani. Collapsing the cardinals of hod. *Journal of Mathematical Logic*, 15(02):1550007, 2015.
- [12] Raffaella Cutolo. Berkeley cardinals and the structure of $L(V_{\delta+1})$. *The Journal of Symbolic Logic*, 83(4):1457–1476, 2018.
- [13] Raffaella Cutolo. The cofinality of the least Berkeley cardinal and the extent of dependent choice. *Mathematical Logic Quarterly*, 65(1):121–126, 2019.
- [14] Raffaella Cutolo. N -Berkeley cardinals and weak extender models. *The Journal of Symbolic Logic*, 85(2):809–816, 2020.
- [15] Carlos A Di Prisco and William S Zwicker. Flipping properties and supercompact cardinals. *Fundamenta Mathematicae*, 109(1):31–36, 1980.
- [16] Matthew Foreman. Some problems in singular cardinals combinatorics. *Notre Dame Journal of Formal Logic*, 46(3):309–322, 2005.
- [17] Kurt Gödel. Remarks before the princeton bicentennial conference on problems in mathematics. 1990.
- [18] Gabriel Goldberg. Choiceless cardinals and the continuum problem. *arXiv preprint arXiv:2201.11557*, 2022.
- [19] Gabriel Goldberg. *The ultrapower axiom*, volume 10. Walter de Gruyter GmbH & Co KG, 2022.
- [20] Gabriel Goldberg. Measurable cardinals and choiceless axioms. *Annals of Pure and Applied Logic*, 175(1):103323, 2024.

- [21] Gabriel Goldberg. Strongly compact cardinals and ordinal definability. *Journal of Mathematical Logic*, 24(01):2250010, 2024.
- [22] Gabriel Goldberg. The uniqueness of elementary embeddings. *The Journal of Symbolic Logic*, 89(4):1430–1454, 2024. doi: 10.1017/jsl.2024.60.
- [23] Gabriel Goldberg and Farmer Schlutzenberg. Periodicity in the cumulative hierarchy. *Journal of the European Mathematical Society*, 26(6):2091–2126, 2023.
- [24] Nai-Chung Hou. Higher structural reflection and very large cardinals. Master’s thesis, University of Barcelona, 2024.
- [25] Ronald Jensen. Inner models and large cardinals. *Bulletin of Symbolic Logic*, 1(4):393–407, 1995.
- [26] Akihiro Kanamori. *The higher infinite: large cardinals in set theory from their beginnings*. Springer Science & Business Media, 2008.
- [27] Peter Koellner. On the question of absolute undecidability. *Philosophia Mathematica*, 14(2):153–188, 2006.
- [28] Kenneth Kunen. *Set Theory: An Introduction to Independence Proofs*. Elsevier, 1980.
- [29] Richard Laver. Implications between strong large cardinal axioms. *Annals of Pure and Applied Logic*, 90(1-3):79–90, 1997.
- [30] Penelope Maddy. Believing the axioms. i. *The Journal of Symbolic Logic*, 53(2):481–511, 1988.
- [31] Paolo Mancosu, Thomas Scanlon, and John Steel. In memoriam: Jack Howard Silver. <https://senate.universityofcalifornia.edu/in-memoriam/files/jack-silver.html>. Accessed: 10 June 2025.
- [32] Donald A Martin and John R Steel. A proof of projective determinacy. *Journal of the American Mathematical Society*, 2(1):71–125, 1989.
- [33] Itay Neeman. Inner models in the region of a woodin limit of woodin cardinals. *Annals of Pure and Applied Logic*, 116(1-3):67–155, 2002.

- [34] Farmer Schlutzenberg. On the consistency of ZF with an elementary embedding from $V_{\lambda+2}$ into $V_{\lambda+2}$. *Journal of Mathematical Logic*, page 2450013, 2024.
- [35] Jack Howard Silver. *Some applications of model theory in set theory*. PhD thesis, University of California, Berkeley, 1966.
- [36] Robert M Solovay, William N Reinhardt, and Akihiro Kanamori. Strong axioms of infinity and elementary embeddings. *Annals of Mathematical Logic*, 13(1): 73–116, 1978.
- [37] William Tait. The iterative hierarchy of sets. *Iyyun: The Jerusalem Philosophical Quarterly*, pages 65–79, 1990.
- [38] William W Tait. Foundations of set theory. *Truth in mathematics*, pages 273–290, 1998.
- [39] William W Tait and Matthias Schirn. Zermelo’s conception of set theory and reflection principles. *The philosophy of mathematics today*, pages 469–483, 1998.
- [40] Hao Wang. *A logical journey: From Gödel to philosophy*. MIT Press, 1997.
- [41] Philip Welch. Global reflection principles. In *Logic, methodology and philosophy of science: proceedings of the fifteenth international congress*. College Publications, 2017.
- [42] Philip Welch and Leon Horsten. Reflecting on absolute infinity. *The Journal of Philosophy*, 113(2):89–111, 2016.
- [43] Philip D Welch. Obtaining woodin’s cardinals. In *Foundations of Mathematics: Logic at Harvard: Essays in Honor of W. Hugh Woodin’s 60th Birthday, March 27-29 2015, Harvard University, Cambridge, MA*, pages 161–176. American Mathematical Society, 2017.
- [44] Hugh W Woodin. The continuum hypothesis. In *Logic Colloquium 2000 (hardcover)*, pages 143–197. AK Peters/CRC Press, 2005.
- [45] W Hugh Woodin. Suitable extender models i. *Journal of Mathematical Logic*, 10(01n02):101–339, 2010.
- [46] W Hugh Woodin. In search of ultimate-1 the 19th midrasha mathematicae lectures. *Bulletin of symbolic logic*, 23(1):1–109, 2017.

- [47] W Hugh Woodin. Large cardinals at the brink. *Annals of Pure and Applied Logic*, 175(1):103328, 2024.